

Smart Air Monitoring and Voice-Assistant Automation System

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This project stands as a testimony to Sir's inspiring mentorship, and we hope it reflects, even in a small measure, the high standards he sets for his students.

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Abstract

This project presents an IoT and TinyML-based smart environmental monitoring and automation system using ESP8266, MQ135, PIR, DHT11, OLED display, Relay-controlled fan, and a laptop-based voice assistant. The system collects sensor data, predicts environmental conditions using a TinyML model, and provides intelligent voice-based suggestions via laptop speaker. It also accepts user voice commands through the laptop microphone to control the fan and LED through the ESP8266.

1 Introduction

1.1 Features to be Developed

- Real-time air quality monitoring using MQ135.
- Temperature and humidity detection using DHT11.
- Motion detection using HC-SR501 PIR (Passive Infrared Sensor).
- Automatic fan activation using Relay + ESP8266.
- OLED screen for live system display.
- Laptop-based voice assistant using Python + TensorFlow Lite.
- Laptop microphone as input, speaker as output.
- ESP8266–Laptop communication using Serial or WiFi.
- TinyML suggestions for improving air quality.

1.2 Drawbacks in Existing Solutions

- No integration of laptop-based offline voice assistants.
- No real-time ML suggestions in low-cost systems.
- Expensive and closed commercial devices.

1.3 Define

Indoor pollution and unmonitored environments create health risks. The challenge is to build a low-cost, AI-enabled, voice-assisted monitoring system accessible to students and institutions.

1.4 Ideate

- ESP8266 for real-time sensor data collection.
- Use TinyML on the laptop for prediction and suggestions.
- Enable laptop-based voice control using microphone input.
- Use relay-controlled fan for air purification.
- Display readings on OLED.

2 Prototype

2.1 Design Explanation

2.1.1 Sensors

- **MQ135:** Detects harmful gases.
- **DHT11:** Measures temperature and humidity.
- **PIR Motion Sensor:** Detects human presence.

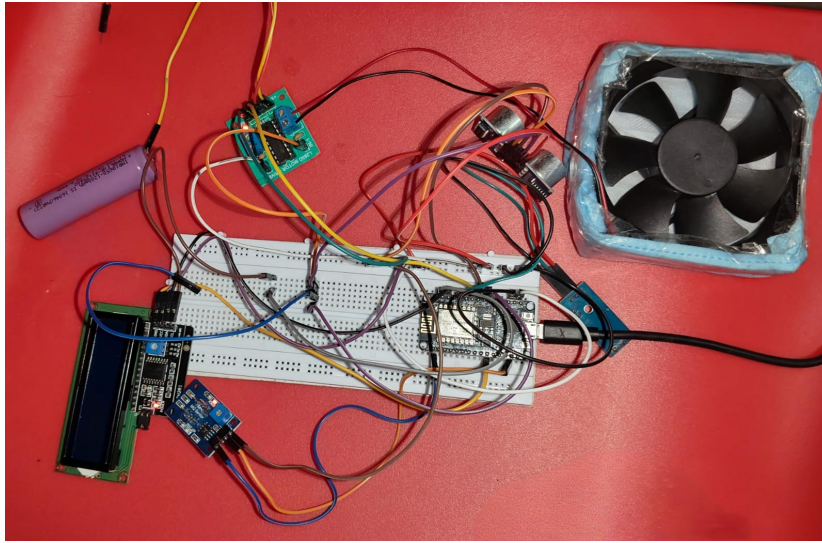


Figure 1: Prototype of the Smart Air Monitoring and Voice Assistant System

2.1.2 Actuators

- **Relay Module + Fan:** Turns fan ON/OFF.
- **LED Indicator:** Shows system status.

2.1.3 Microcontroller

- **ESP8266 NodeMCU:** Reads sensors, sends data to laptop, receives commands.

2.1.4 Software Platforms Used

- **Python + TensorFlow Lite:** Runs TinyML model, processes audio commands.
- **SpeechRecognition + PyAudio:** Converts microphone input to text.
- **PySerial:** Sends fan/LED commands to ESP8266.
- **Arduino IDE:** ESP8266 programming.

3 Implementation of Code

3.1 ESP8266 Arduino Code

```
#include <ESP8266WiFi.h>
#include <DHT.h>
#include <Wire.h>
#include <Adafruit_SSD1306.h>
#define DHTPIN 5
```

```

#define DHTTYPE DHT11
#define MQ135 A0
#define PIR 14
#define RELAY 12
#define LED 13
DHT dht(DHTPIN, DHTTYPE);
Adafruit_SSD1306 display(128, 64, &Wire, -1);
void setup() {
    Serial.begin(9600);
    pinMode(RELAY, OUTPUT);
    pinMode(LED, OUTPUT);
    pinMode(PIR, INPUT);
    dht.begin();
    display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
}
void loop() {
    float t = dht.readTemperature();
    float h = dht.readHumidity();
    int gas = analogRead(MQ135);
    int motion = digitalRead(PIR);
    // Send data to laptop
    Serial.printf("%0.1f,%0.1f,%d,%d\n", t, h, gas, motion);
    // Receive command from laptop
    if (Serial.available()) {
        char c = Serial.read();
        if (c == 'F') digitalWrite(RELAY, HIGH); // Fan ON
        if (c == 'f') digitalWrite(RELAY, LOW); // Fan OFF
        if (c == 'L') digitalWrite(LED, HIGH); // LED ON
        if (c == 'l') digitalWrite(LED, LOW); // LED OFF
    }
    delay(200);
}

```

3.2 Python TinyML + Voice Assistant Code

```

import serial
import speech_recognition as sr
import tflite_runtime.interpreter as tflite
import numpy as np
import pytttsx3
ser = serial.Serial('COM5', 9600)
engine = pytttsx3.init()
# LOAD TINYML MODEL
interpreter = tflite.Interpreter(model_path="model.tflite")
interpreter.allocate_tensors()
def predict(values):
    input_details = interpreter.get_input_details()
    output_details = interpreter.get_output_details()
    interpreter.set_tensor(input_details[0]['index'], np.array([values], dtype=

```

```

        interpreter.invoke()
        return interpreter.get_tensor(output_details[0]['index'])[0]
r = sr.Recognizer()
while True:
    data = ser.readline().decode().strip().split(",")
    t, h, gas, motion = map(float, data)
    result = predict([t, h, gas, motion])
    if result > 0.7:
        engine.say("Warning: Air quality is bad. Please turn on the fan.")
        engine.runAndWait()
    with sr.Microphone() as mic:
        print("Listening...")
        audio = r.listen(mic)
        text = r.recognize_google(audio)
        if "fan on" in text.lower():
            ser.write(b'F')
            engine.say("Turning on the fan.")
            engine.runAndWait()
        if "fan off" in text.lower():
            ser.write(b'f')
            engine.say("Turning off the fan.")
            engine.runAndWait()

```

4 Test and Results

System successfully:

- Recorded sensor data.
- Sent it to the laptop.
- Generated TinyML-based suggestions.
- Accepted voice commands.
- Controlled fan and LED via ESP8266.

5 Future Scope

- Android app integration.
- Larger TinyML models.
- Cloud-based storage.